

Essentialism and the Essence of to be Necessary

In his 1994 paper *Essence and Modality* Kit Fine argued against the view that essence should be understood in terms of *de re* modality. He suggested instead to understand metaphysical necessity by essence of things. The idea to explain, especially to ground modality in essence was followed by various philosophers albeit with some differences in details. I term all such accounts *essentialism* on modality.

To ground modality on essence, essentialists have to provide an account on essence that does not rely on modal notions. And at the same time, they must also derive the necessity of p from the essentiality of p . Some philosophers argue that this has never been successfully established. The arguments provided for the latter implicitly assume some modal characteristic on the essentiality of p , which is to fail to satisfy the former requirement. And if the essentiality of p has no modal import at all, there seems to be a gap between what it is to be essential and what it is to be necessary.

In this paper, I will argue that essentialists can reply to the challenge of gap without compromising their core theses. One remaining possibility is overlooked: the gap is bridged by the essence of *to be necessary*. More specifically, I argue that both requirements can be satisfied by assuming that it is essential of to be necessary that if something is essentially true, then it is necessarily so.

In section 1, I will introduce the problem of the source of necessity, and argue all essentialist accounts give non-transmissive explanation to necessity, even if it was not explicitly stated. The challenge of gap will be elaborated in detail in section 2. I argue in section 3 and 4 that what the challenge actually shows is that any account on the source of necessity that utilizes non-transmissive explanation must involve certain commitment to the essence of to be necessary. It would be incorrect to conclude from the challenge that essentialism is untenable. In section 5, I will clarify why my suggestion is legitimate. Especially, I argue that it does not face the problem of circularity. Note that I am not trying

to argue for the essentialist answer for the problem of the source of necessity, against other alternatives.

1 The Source of Necessity and Non-Transmissive Explanations

For any necessary truth p , we can ask *why* is p necessary. It is different from the question of why p is true. For example, we can answer the question of why is $(3^2 = 9 \vee \text{snow is white})$ true by saying that $3^2 = 9$ is true. But this cannot be the answer for the question of why the disjunction is necessary.

Similarly, the question of the source of a *particular* necessity should be distinguished from the question of the *general* source of necessity. The particular necessity of the previous disjunction might be explained by the necessity of $3^2 = 9$. This however does not guide us at all about the general source of necessity, since the necessity of $3^2 = 9$ is left unexplained.

Blackburn (1993) argued that the general source of necessity cannot be given by some true propositions. Suppose we want to claim that $\Box p$ because q . If q is necessary it cannot be the final source of necessity, for it leaves the necessity of q unexplained. But q also cannot be contingent, for “... then there is strong pressure to feel that the original necessity has not been explained or identified, so much as undermined.” (Blackburn 1993, p.53)

Blackburn wishes to conclude that therefore any realist conception on necessity fails. This advance might be too hasty (Cameron, 2010). However, it was also argued that it is indeed possible to explain necessity via contingent or necessary truths. Hale (2002) pointed out that denying the contingency horn presupposes that necessity is never contingent, which some philosophers, notably conventionalists would disagree. But Hale himself took the other option. He argued that we can explain necessity by necessary truths, insofar as the explanation is *non-transmissive*. A transmissive explanation takes the form of $\Box p$ because $\Box q$, where q itself cannot explain $\Box p$. Perhaps Blackburn is correct if this is the only available form of explanation via necessary truths. A general scheme for giving transmissive explanations may help us to sort out so-called “fundamental” necessities from non-fundamental ones, but does not offer a theory of general source of necessity.

Non-transmissive explanations, by contrast, take the form of $\Box p$ because q , where q is necessarily true “but in which what explains the necessity of the explanandum is not q ’s *necessity*, but its *truth simpliciter*.” (Hale 2002, p.311)

Here the regress does not occur. Since q is also a necessary truth, the necessity of $\Box q$ would also be explained via the truth of a proposition which is necessarily true, say r . But that does not imply that the necessity of p is thus explained by r . For, it is the truth *simpliciter* of q that explains the necessity of p , not the necessity of q .

Therefore, a general scheme of giving non-transmissive explanation to all, or at least every fundamental necessities suffices as an account for the general source of necessity. This is typically done by specifying what condition must an arbitrary p satisfy to be necessary. Hale’s preferred view is that “Necessarily p because $\Box_{X_1, \dots, X_n} p$ ” (Hale 2013, p.152): the necessity of any proposition p is explained by it being essential to some objects $X_1, \dots, X_n$.¹ There are other alternatives. For instance, possible world realists would assert that $\Box p$ because p is true in all possible worlds. While it would be necessary that p is true in all possible worlds, it is not the necessity doing the explanatory work. I will classify all those theories of the source of necessity which utilizes non-transmissive explanation as “non-transmissivism”.

Before proceeding, let me clarify myself between various senses and kinds of essence. By “essence” I mean *mediate and consequential* essence, following Fine (1995a). Suppose we have a clear idea of a strict and direct essence, which is immediate and constitutive. Then, mediate essence of F includes both the nature of itself and the nature of things that are involved in the nature of F ; consequential essence of F includes what logically follows from the nature of F .

Also crucial is the distinction between objectual essence and predicational essence. On the one side there is the essence of a particular object, and the other side the essence of to be such and such. Fine (2015) argued the essence of to be F is different from the essence of the concept F . Suppose the objectual essence of the concept *concept* includes the fact that *concept* is a concept. Then it is a consequential essence of it that $\exists x(x \text{ is a concept})$. However, it seems quite implausible to think that it is by nature of *what it is to be a concept* that there is something satisfies it. That is, while $\exists x(x \text{ is a concept})$ is plausibly an essence of the concept *concept*, it would not be an essence of to be a concept. This, we may say, illustrates the difference between the objectual essence of concept and predicational essence of to be a concept.

It would be clear that the essence of objects or things would be the objectual essence of that object. And following Fine (1995a, 2015), by $E_F p$ ² I mean

¹I use two terms “true in virtue of the nature of”, “be an essence of” and “essential to” interchangeably, following Fine and Hale.

²I use “ E ” as the essentialist operator, substituting Fine (1995a)’s notation $\Box$. “ $\Box$ ” is exclu-

p is a predication essence of F . Objectual essence will be notated by using the *rigid predicate* $\lambda x.x = a$, where “ a ” is a name for an object. To say p is an objectual essence of a I write “ $E_{\lambda x.x=a}p$ ”. I will occasionally write E_a to denote the objectual essence of a , but this is merely an abbreviation of $E_{\lambda x.x=a}$.

Note that there can be slightly different views among essentialists on the relation between essence and necessity. For Kit Fine, to be metaphysically necessary is *defined* as to be essential to *all objects* (Fine 1994, 1995a). Therefore, although not explicitly stated, Fine would agree that the necessity of p is grounded on p ’s being essential to all objects. Following Fine (2015), such a definition which gives the constitutive necessary and sufficient condition for to be necessary yields, that to be true in virtue of the nature of all objects is both (partial) ground and essence of to be necessary. By contrast, Hale explicitly claims his account to be non-reductive. Although he agrees with Fine that metaphysical necessity is grounded in the nature of things, he also takes modality to be “fundamental and irreducible” (Hale 2013, Ch.3).

However, regardless of the subtle difference, any sensible essentialist account on modality is also non-transmissivist. Since essentialists hold that the necessity of p is explainable by it being *objectually* essential to something,³ they are automatically committed to the position that the source of necessity is also necessary. Under the standard axioms of the logic of essence, it is a theorem that $E_F p \rightarrow E_F E_F p$, where the predicate F is rigid (Fine 1995a, p.255, see THEOREM 6 (ii)). That is, if p is essential to some object, the fact that p is essential to the object is essential to the object in question itself. Since essentialists should accept that p is necessary if p is essential to some objects (this will be argued shortly), $E_F p$ which is the source of $\Box p$ must also be necessary; and the source of $\Box E_F p$ would be $E_F E_F p$, which is also necessary. And so on. Also remind that essentialists suggest $E_F p$, not the necessity of $E_F p$ to be the explanans of $\Box p$. Thus, essentialists turn out to be holding that the source of necessity is necessarily true yet what does the explanatory work is its *truth simpliciter*, not its necessity. And that is exactly what non-transmissivism is.

sively used as a necessity operator.

³Fine (1994): “The metaphysically necessary truths can then be identified with the propositions which are true in virtue of the nature of all *objects* whatever” (p.9, italics added); Hale (2013): “Metaphysical necessities might plausibly be identified as those which hold in virtue of the nature of *things* in general.” (p.150, italics added).

2 The Gap between Essence and to be Necessary

One constraint for all essentialist accounts is that the notion of to be essential should not be understood in terms of to be necessary. For example, suppose for Fa to be the essence of a just is for a to necessarily have F . Then it is unsatisfactory to say the necessity of Fa is explained by Fa being essentially true. It just amounts to say Fa is necessary because a necessary has F , and that leaves the *de re* necessity unexplained. It is parallel to the reason why possible world realists should not define to be a possible world as to be a world that is metaphysically possible. But then, how can essentialists guarantee that all essential facts are necessary?

All essentialists must accept that all objectually essential truths are necessary. To acknowledge contingent objectual essences, one must give a criterion demarcating them from necessary objectual essences without appealing to the notion of essence. That is, one must identify the non-essentialist source of necessity, which contingent essences lack. And this apparently undermines essentialism.

Given such a constraint, some philosophers question how can essentialists guarantee all essential truths are necessary. To put it formally, essentialists are committed to what Mackie (2020) calls “Necessity Principle (NP)”:

$$(NP) \quad \forall F \forall p (E_F p \rightarrow \Box p)$$

where F is rigid.⁴ I will occasionally omit one or two universal quantifiers for convenience and readability.

One common understanding of essence among essentialists is by *real definition*: roughly, the essence of certain thing or to be such and such is captured by their real definition. (Fine 1994, p.2; Hale 2013, p.132; Lowe 2012, p.919). Alternatively, Correia and Skiles (2017) suggested an understanding in terms of generalized identity, expressed by the identity statements such as “for a thing to be a bachelor is for it to be an unmarried adult male” or “for the Atlantic Ocean to be filled with water is for it to be filled with H_2O molecules” (p.643). Here to be F expresses the full essence of to be G if and only if to be G is to be F ; it expresses partial essence of to be G if and only if there is some H such that to be G is to be both F and H (pp.649-50)⁵. The charge is that it does not seem

⁴This formalization follows Fine (2020).

⁵Here again the difference between objectual and predicational essence will be captured by the difference between rigid predicates and non-rigid predicates. Suppose the *Con* is the predicate *to be a concept*. And using $\equiv$ as the generalized identity operator, $Con(x) \equiv_x p$ states

that the nature of real definition or generalized identity can ground (NP) alone. How can essentialists “deliver a modal rabbit out of a non-modal hat” (Mackie 2020, p.252)?

Hale tried to give an argument. Suppose $\Phi\alpha$ is an objectual essence of α . Now suppose it is possible that $\neg\Phi\alpha$. It implies that there is a β such that $\alpha = \beta \wedge \neg\Box\Phi\beta$. But β cannot be identical to α in the world such that $\neg\Phi\beta$. For, it does not satisfy the condition to be identical to α , which is Φ . Therefore by the necessity of identity $\neg\alpha = \beta$, contradiction. Since contradictions are impossible, we can conclude that it is not possible that $\neg\Phi\alpha$. In other words, if $E_{\lambda x.x=\alpha}\Phi\alpha$, then $\Box\Phi\alpha$ (Hale 2013, p.133).

But it has been pointed out that the argument does not successfully address the challenge (Romero 2019, pp.125-7; Leech 2021, pp.893-5). For the proposed argument to succeed, it must be assumed that Φ is *necessarily* the condition for to be identical to α . Either $\Box E_\alpha\Phi$ or $\Box\Phi\alpha$ is required. But what is assumed is only that Φ is *actually* the condition of the identity. To hold that the assumption implies $\Box\Phi\alpha$ simply begs the question. And while $E_\vee(E_{\lambda x.x=\alpha}\Phi)$ ⁶ is provable from $E_{\lambda x.x=\alpha}\Phi$, the inference by Finean definition from $E_\vee p$ to $\Box p$ is problematic, as we will see below. Moreover, Hale himself would not agree to the definition.

In fact, one can construct a logical proof for (NP) once she accepts the Finean definition of necessity. It is an axiom in the logic of essence that $F \subseteq G \rightarrow (E_F p \rightarrow E_G p)$, where $R \subseteq G$ is for $\Pi x(Rx \rightarrow Gx)$ (Fine 1995a, pp.246-7), and Π is a quantifier ranging everything in the domain. Since for any rigid predicate $F \subseteq \vee$, it follows that $E_F p \rightarrow E_\vee p$. And since $\Box p$ is just defined as $E_\vee p$, we can derive $E_F p \rightarrow \Box p$, which is (NP) itself. But there is another reason to regard the definition to be unsatisfactory (Romero 2019, p.129-36). As it was mentioned, Fine defines necessary truths as propositions that are true by the essence of all objects. However, by “all objects” he means all objects in the domain, which is the set of all possible objects. Persuasively, since the problem of which objects are possible depends on the notion of metaphysical modality, to conceive necessary truths the essence of all *possible* objects is circular. In the same paper Fine also writes “[...] one might operate with a broader conception of possible object than is customary in modal logic” (Fine 1995a, p.244), but it is still unclear what the “broader conception of possible object” is.

To understand essence in terms of generalized identity may seem to offer

that p is the full predicational essence of to be a concept; $[\lambda x.x = \text{Con}]y \equiv_y q$ states that q is the full objectual essence of the concept *concept*.

⁶ $E_\vee p$ may be read: p is an essence of all objects. There are logical machinery behind it in Fine (1995a), but it would be irrelevant here, and I do not want to overcomplicate things.

better prospects for advocating (NP), since “[a]s with objectual identity, generalized identity is tightly linked to metaphysical necessity” (Correia and Skiles 2017, p.656). Following Correia and Skiles, let “ $p \equiv q$ ” denote the generalized identity between p and q . Here $p \equiv p$ is a theorem⁷ and therefore in any normal modal logic it follows that $\Box(p \equiv p)$. And given $p \equiv (q \wedge r)$, which means that q is a partial essence of p , by substitutability we get $\Box(p \equiv (q \wedge r))$. That is, assuming q is a partial essence of p , it follows that q is a partial essence of p necessarily.⁸

This actually proves something stronger than (NP): namely, $E_F A \rightarrow \Box E_F A$. However, since anything true in virtue of the nature of some object would be true *simpliciter*, this stronger consequence implies (NP). The problem here is that the proof presupposes $\Box(p \equiv p)$: that the generalized identities are necessary. What is the source of the necessity of $(p \equiv p)$? It has been argued that the necessity of $\Box(p \equiv p)$ cannot be explained by the essence of generalized identity (Leech 2021, pp.901-2; cf. Romero 2019, p.138). For example, suppose that the necessity is explained by that *for two things to be generically identical is partially to be in a reflexive relation*. Grant, but what explains the necessity of the partial generalized identity between generalized identity and reflexive relation? Here we cannot appeal to the necessity of $p \equiv p$, for that exactly is the supposed explanandum. Therefore, to appeal to the essence of generalized identity itself does not solve the problem.

Leech seems to conclude that in explaining the necessity of the generalized identity “one should not appeal to the essence of anything to account for this necessity”. She further considers this constraint as “the crucial and general objection” on the necessity of essence (p. 902).

I disagree. I think Leech overlooks another possibility: that generalized identities are necessary because of the essence of *to be necessary*. Furthermore, in my view, this also applies directly to (NP). Every truth essential to something or all things are also necessarily true, not because of the nature of essence but because of the nature of what it is to be necessary. In remaining sections, I will argue that the solution for essentialists is to locate (NP) as a partial essence of to be necessary - the thesis I will call (NPEN).

⁷For they take generalized identity to be reflexive (p.645)

⁸It would be clear that in Correia and Skiles’ definition the full essence is trivially also a partial essence.

3 Non-transmissivism and the Essence of to be Necessary

I argued in section 1 that all essentialists should be non-transmissivists. Typically, in a non-transmissive explanation $\Box p$ because q , the role of q is taken by a proposition *about* p . An example would be that p is true in all possible worlds. Let C be any such status or criterion a proposition can satisfy. A specific non-transmissivist theory on the general source of necessity will then give an account of what the status C is. Thus, non-transmissivists would generally hold that $\Box p$ because $C(p)$, where p necessarily satisfies C . Essentialists are those who hold that the status C is *to be objectively essential*.

All non-transmissivists should affirm the General Necessity Principle (GNP): $\forall p(C(p) \rightarrow \Box p)$. The reason is same as the reason why essentialists must accept (NP). If there are some true propositions that satisfy C but are not necessarily true, it means that there are something additional to C that makes propositions satisfying C necessary. C is then at most only a part of the criteria a proposition must satisfy to be necessary, contrary to the initial assumption.

This indicates that the problem of gap is not peculiar to essentialists. It turns out to be a general difficulty for any non-transmissivist account. For example, a possible world realist would give an explanation: $\Box p$ because p is true in all possible worlds. But if the notion of possible worlds is defined without modal notions, one can equally ask, following Mackie, how possible world realists can “deliver a modal rabbit out of a non-modal hat”. The general difficulty is to provide an elucidation of how a non-modal understanding of C can get along with (GNP) well.

How can a non-transmissivist guarantee (GNP)? First, one *cannot* merely appeal to some fact about C or necessity that is not essential of to be C or to be necessary, but take $C(p)$ to be the ultimate source of $\Box p$. Suppose a proposition satisfying C is necessary as long as C satisfies other non-essential criteria, say X . That is, (GNP) depends on $X(C)$, but $X(C)$ is not a partial essence of to be C nor to be necessary. Then, what *explains* $\Box p$ is not merely $C(p)$ but $C(p) \wedge X(C)$, since $C(p)$ cannot guarantee $\Box p$ alone. Additional component $X(C)$ is required because the nature of to be C or to be necessary does not by itself ensure $X(C)$. So the non-transmissivist account that $\Box p$ because $C(p)$ fails, for to satisfy C turns out not to be everything that is needed to explain necessity.

Similarly, we cannot appeal to non-essential property of to be necessary.

Suppose (GNP) obtains as long as $Y(\Box)$ obtains⁹; and that is neither an essence of to satisfy C nor to be necessary. The correct account then should be that $\Box p$ because $C(p) \wedge Y(\Box)$, and the original account turns out to be untenable. To sum up, if one wants to insist that $C(p)$ provides full explanation for $\Box p$, one cannot think (GNP) obtains regardless of the nature of to satisfy C or to be $\Box$.

The previous argument does not show it cannot be the case that $X(C)$ or $Y(\Box)$ grounds (GNP). It shows that even if one believes that $X(C)$ or $Y(\Box)$ grounds (GNP), in order to be a non-transmissivist, one should also think the essence of to be C or to be necessary is sufficient to guarantee (GNP). However, non-transmissivists also cannot take (GNP) to be a constitutive essence of to satisfy C . Suppose for the contrary that $E_C(C(p) \rightarrow \Box p)$. Now, if one takes the notion of essence as closely tied to the notion of real definition, it follows that to satisfy C is partially *defined* in a way such that it is the case that $C(p) \rightarrow \Box p$. Or following a revised version of definitional essentialism (Fine 2015), it follows that $\Box p$ is constitutive necessary condition for some p to satisfy C . It seems to me, then, to satisfy C depends partially on to be necessary, and therefore $C(p)$ cannot ground $\Box p$.

This can also be effectively shown in terms of generalized identity. Suppose for the contrary that $C(p) \equiv_p \Box p \wedge q$: to be C just is to be necessary and q . Now, since a statement of generalized identity closes a theoretical gap (Rayo, 2013); if the left side of a generalized identity statement is the explanans of a metaphysical explanation, the right side must be able to substitute it. A brief example: suppose the x 's not being the subject of the marriage loan is metaphysically explained by x 's being a bachelor. And since to be a bachelor just is to be an unmarried adult male, x 's not being the subject of the marriage loan would also be metaphysically explained by x 's being an unmarried adult male. Similarly, given $C(p) \equiv_p \Box p \wedge q$, the claim that $C(p)$ metaphysically explains $\Box p$ is equivalent to the claim that $\Box p \wedge q$ metaphysically explains $\Box p$. And that apparently does not seem satisfactory, therefore the claim that $C(p)$ metaphysically explains $\Box p$ should be discarded.¹⁰

Perhaps the argument can be made clearer by introducing Fine's notion of ontological dependence. According to his account on ontological dependence (Fine 1995b), x depends on y "if y is a constituent of a proposition that is true in virtue of the identity of x " (p.275). Therefore from $E_C(C(p) \rightarrow \Box p)$, it follows that to be C ontologically depends on to be necessary. Of course Finean ontological dependency is not antisymmetric. However, what non-transmissivists

⁹Here I take the operator $\Box$ to be a constant higher-order predicate which is equivalent to "to be necessary" in natural language.

¹⁰A similar argument is anticipated by Romero (2023)

need is that to be C is strictly prior than to be necessary - which prevents to satisfy C being defined, or constituted in terms of to be necessary.

The only remaining possibility, then, is to say that (GNP) is an essence of to be necessary. Any non-transmissivist account on necessity must take their relevant instances of (GNP) to be true because of the essence of to be necessary. And in fact Fine's account implies that, although not explicitly stated. As I mentioned earlier, Fine *defines* $\Box$ in terms of essence: $E_{\forall}p \leftrightarrow_p \Box p$. And its implication $E_{\forall}p \leftarrow_p \Box p$ is just another, intuitive way to put $E_{\Box}\forall p(E_{\forall}p \rightarrow \Box p)$ (Fine 2015, p.299). I also mentioned that for any rigid predicate F $E_Fp \rightarrow \Box_{\forall}p$. Since from $E_{\forall}p \rightarrow \Box p$ and $E_Fp \rightarrow \Box_{\forall}p$ it follows that $E_Fp \rightarrow \Box p$; we can get $E_{\Box}\forall F\forall p(E_Fp \rightarrow \Box p)$.

But I also mentioned that Finean definition is accused to be circular, for he presupposes the domain of *all metaphysically possible* objects. An additional concern is that we might want to talk about the essence of *impossible* objects, which are not in any broad sense of "possible object" one of them. Typical examples is the Russellian set. Why is the Russellian set impossible? An essentialist might want to argue as follows. The Russellian set is essentially a set of all sets that does not contain itself; therefore it is necessary that for every x , if x is identical to the Russellian set, it is a set of all sets that does not contain itself. And it is also necessary that, there is no x such that it is the set of all sets that does not contain itself. Therefore, Russellian set necessarily does not exist, i.e., is an impossible object. But for such an argument to be sound, it should be held that the essence of an impossible object can also be the source of some necessities.

But then how could the Finean definition be maintained? One way out would be to construct a domain which includes both possible and impossible objects. But which objects, possible or impossible, are there? Perhaps Meinongians could give us an answer.¹¹ However, I think it is an unnecessary detour to think that essentialists would need a theory of objects.¹² It is not very desirable for essentialists to just wait until object theorists come up with a comprehension principle of objects everyone can happily endorse.

¹¹See, for example Zalta (1983) or Priest (2005). Fine himself also worked on the problem of non-existent and the comprehension principle on objects. The negative part, which he criticizes certain kinds of object theories, was published in Fine (1982). There he also stated that the positive project where he develops his own theory will be published as the part II of the paper, but unfortunately it remains as a non-existent object till now.

¹²It is even unclear if an independently developed object theory would get along with essentialism well. For example, Zalta, although I see no logical inconsistency between his theory of objects and essentialism, advocates a modal conception of essence specified in terms of his theory of objects in Zalta (2006).

Instead, I suggest discarding the Finean definition, and take (NPEN)¹³:

$$(NPEN) \ E_{\Box} \forall F \forall p (E_F p \rightarrow \Box p)$$

not as a consequence of the definition, but directly into the essentialist account on necessity. Here again F range Only over rigid properties. If we follow definitional conception on essence, (NPEN) amounts to say that (NP) is part of the definition of *to be necessary*, not the definition of *to be objectually essential*. Still the Finean intuition would be preserved under any suitable theory of objects, as long as the axiom $F \subseteq G \rightarrow (\Box_F p \rightarrow \Box_G p)$ is employed. Of course this generalizes. For any non-transmissivists, the generalized version - (GNPEN): $E_{\Box} \forall p (C(p) \rightarrow \Box p)$ - must be taken into account.

Since any essentialism should be non-transmissivist, all essentialists must accept (NPEN). Otherwise, either essence turns out not to provide a full explanation for necessity; or essence cannot ground necessity because it infringes priority or irreflexive constraint of ground. One consequence is that any compelling argument for essentialism would also be an argument for (NPEN). For I do not aim in this paper to argue for the essentialist answer against other alternatives, I will not explore possible specific arguments for (NPEN).

4 How to Bridge the Gap

What I *should* show in this paper, is how essentialists can answer the challenge of gap by utilizing (NPEN). To briefly reiterate, their main criticism was that once we stick to the notion of essence that is *not* defined in terms of necessity, there seems to be a gap between essence and necessity. My suggestion is, of course: the gap is to be bridged by the essence of to be necessary.

Let us return to Hale's argument for the necessity of definitional essence. I take Romero's criticism to be partially correct, to the extent that we cannot infer $\Box \Phi \alpha$ from $E_{\alpha} \Phi \alpha$ without appealing to the essence of to be necessary. The gap, however, can be bridged by the nature of to be necessary. So the revised argument goes as follows. Suppose again, that $E_{\alpha} \Phi \alpha$. Now, Romero and Leech ask: why β such that $\neg \Box \Phi \beta$ cannot be identical to α , if the initial presupposition was only that it is *actually* $E_{\alpha} \Phi \alpha$? Now we have a further reply for that. Suppose for contradiction that $\alpha = \beta$ and therefore by substitutability $\neg \Box \Phi \alpha$.

¹³Necessity Principle as the Essence of to be Necessary

Now, from (NPEN) we can prove, utilizing the *T-axiom*¹⁴ and modus ponens that $\forall F \forall A (E_F A \rightarrow \Box A)$. By universal instantiation we get $E_\alpha \Phi \alpha \rightarrow \Box \Phi \alpha$; and by the assumption $E_\alpha \Phi \alpha$ and modus ponens we get $\Box \Phi \alpha$. Contradiction. Therefore, there can be no β such that $\alpha = \beta \wedge \neg \Box \Phi \beta$, which ensures that if Φ is actually an essential property of α , then it is a metaphysically necessary condition for something to be identical to α .

It is important to notice that the argument depends on the essence of to be necessary, and the logical rules of the essentialist operator, the universal quantifier and the material conditional. So we do not suppose objectual essences are *ab initio* necessary. The necessity is derived from the essence of to be necessary and the relevant logical rules¹⁵. This fact will be used later in section 5.

This is nevertheless not to ignore the essentialist intuition that it is in some sense “obvious” that objectual essences are necessary. I suggested that essentialists cannot take objectual essence to be *ab initio* necessary. However, it seems also true that there certainly is some kind of frustration when essentialists are confronted with questions like “why are facts of objectual essential necessary?” or “why an object necessarily has the property that defines what the object is?”

This is, I suggest, same kind of frustration when one is confronted with questions like “why is water H_2O ?” or “why does Socrates’ singleton set have Socrates as member?”. According to Glazier (2017), these questions are subject to essentialist explanation.¹⁶ We can simply answer: “that is just what water is” or “that is just what the set is”. Essentialists therefore can take (NPEN) as an explanans of such an essentialist explanation of (NP). “Why are objectual essences necessary?” I suggest one kind of answer essentialists can give is: “that is just what to be necessary is!”

But couldn’t one instead appeal to a different essentialist explanation: objectual essence are necessary because that is just what *to be objectually essential* is? This is similar to the account of Wilsch (2017) or Wallner and Vaidya (2020). Wilsch directly takes (NP) to be the essence of essence, whereas Wallner and Vaidya takes both $E_x p \rightarrow \Box E_x p$ and $E_x p$ can explain $\Box p$ essential to

¹⁴ $E_F p \rightarrow p$ (Fine 1995a, pp.247-8)

¹⁵Which is common to expect to be (objectually) essential to the relevant logical operators. For example, Fine (1994) writes “the logical necessities can be taken to be the propositions which are true in virtue of the nature of all logical concepts” (also see Correia (2020)). At least every instance of the logical rules would be essential to the logical concepts, and even the rule itself if it can be taken as a higher-order proposition.

¹⁶Glazier restricts himself into immediate and constitutive essence. However, his account on essentialist explanation would be able to naturally expand to the mediate essence except the point that essentialist explanation is ultimate.

the essentialist operator E_x . Since $E_x p \rightarrow \Box E_x p$ entails (NP), it would be fair to say both takes (NP) as a mediate and consequential essence - which I simply say ‘essence’ - of to be objectually essential.

This is of course an instance of taking (GNP) the essence of to be C . It makes to be objectually essential no more fundamental than to be necessary. In case of Wilsch, he thus explicitly states himself as a modal primitivist, not an essentialist. Although he takes essence to give the “source” of metaphysical necessity, here what the “sources” actually do is to determine the extension of various kinds of necessity, and account for the naturalness of them. “The sources of necessity are the modal joints of nature.” (Wilsch 2017, p.429). Essence, then, explains why metaphysical necessity is a significant kind of primitive necessity, why it “carves the nature at its joints”. That is apparently a different enterprise from the problem of source Hale has in mind, and what is being discussed in this paper.¹⁷

Yet Wallner and Vaidya take themselves as *non-reductive Finean essentialists*, “which endorse [...] that modality is grounded in definitional essences.” (Wallner and Vaidya 2020, p. 431). They also make it explicit that they intend to follow Hale in giving non-transmissive explanation for necessity. Then, I think, they face the same problem occurring when a non-transmissivist takes (GNP) as an essence of to be C . My argument in section 3 shows that the authors turn out to be committed to the stance that not only necessity is not reduced into essence, but that to be necessary partially constitutes what is to be objectually essential, or at least essence is not prior than necessity. And that seems to be in tension with the claim that modality is grounded in essence, according to some standard understanding of ground (cf. Schaffer 2009, Fine 2012). Also, if one is sympathetic with Correia and Skiles, one can further criticize the account for offering a trivial explanation on necessity, in the form $\Box p$ because $\Box p$ and q .

So much for the definitional conception of essence. If essence is understood in terms of generalized identity, we can translate the same idea into $\exists q(\Box p \equiv_p (\forall x \exists r([\lambda y.y = x]z \equiv_z (p \wedge r))) \wedge q)$. In short: to be necessary *just is* to be a partial essence of some object. We can now proceed similarly as we did in the case of definitional essence. How can we argue here that therefore all objectually essential facts are necessary? Suppose we have a statement of objectual essence: $[\lambda y.y = a]x \equiv_x (s \wedge w)$. s is a partial essence of the object a .

¹⁷My argument in section 3 does show that Wilsch cannot be a non-transmissivist. But that is somewhat trivial, for a modal primitivist would deny there is a “source” of necessity in our sense. Wilsch’s account is best conceived as a complete rival for essentialism - not a variant of it - just as deflationism or dispositionalism is. And to advocate essentialism among a variety of theories would exceed the scope of this paper.

And it follows from the nature of to be necessary that for s to be necessary is just to be a partial essence of a . Since s is a partial essence of a , we get that s is necessary.

Interestingly, here we have another option. Recall that in section 2, the charge was not on (NP), but on $\forall p \Box(p \equiv p)$. Can we account for this stronger thesis than (NP) instead? I think yes. Since we cannot say $\Box(p \equiv p)$ is true by the essence of the generalized identity, the core idea again is that it has something to do with the essence of to be necessary.

But we still should be careful with the formulation. For example, $\exists q(Nx \equiv_x \forall p(\Box(p \equiv p) \wedge q))$ might seem natural; but it would not work. For one thing, the supposition is consistent with there being no necessary truths yet there being true instances of generalized identity. It is also quite mysterious why should the nature of a particular fact having necessity involve that any fact of generalized identity is necessary.

Instead, I suggest: $\exists q(([\lambda G.Gx \equiv_x \Box x]F) \equiv_F (\forall p \Box(p \equiv p) \wedge q))$. This, however, is not to say it is the predication essence of to be necessary that $\Box(p \equiv p)$. It states that it is the partial objectual essence of necessity operator that $\Box(p \equiv p)$. In this case, it is not involved in the essence of some particular proposition to be necessary that any fact of generalized identity is necessary. $\Box(p \equiv p)$ is also guaranteed, for $\Box p \equiv_p \Box p$ is a theorem. Now essentialists can proceed their argument as it was suggested in section 2. We can by substitutability get $\Box(p \equiv (q \wedge r))$ and $\Box(p \equiv q)$, which means that the fact on what is essential to what is necessary.

5 Circularity

I argued in the previous section that the only viable option for non-transmissivists is to take the relevant instances of (GNP) to be a partial essence of to be necessary. This was done by showing that any other alternative is incompatible with non-transmissivism. But to say that it is the only viable option for non-transmissivists is not to say that it is tenable. After all, non-transmissivism might also be incompatible with (GNPEN); and with the argument I presented it completes a trilemma which likely undermines any non-transmissivist theory.

It seems that (GNPEN) is not *logically* inconsistent with non-transmissivism. To say the nature of to be necessary is such and such is one thing, and to say the necessity is explained by such and such factor is another. Similar to essentialism, which is an instance of non-transmissivism.

But it could be the case that the thesis seriously undermines the theoretical motivation of non-transmissivism. The reason to reject (GNP) being essential to *C* was not logical inconsistency. The argument against it was that it results in the disturbing consequences regarding fundamentality or metaphysical explanation.

(NPEN) would not face the problem of fundamentality. It implies that to be objectually essential is more fundamental than to be necessary. But perhaps there are other problems. Romero (2019) states six constraints on the notion of explanation which the ‘explanation’ of modality by essence must satisfy:

Why The explanation must answer to a ‘why’ question.

Acyclicity The explanation should not be circular.

Fundamentality The explanans should be metaphysically more fundamental than the explanandum.

Ontic and Factive It must be the case that the truth of the explanans, not our concept or theory of it, explains the truth of the explanandum.

Structural The explanandum need not be an object or a property.

Objectivity The explanation should be objectively valid. (p.124)

It is already shown that (NPEN) does not make essentialism infringe **Fundamentality**. Essentialism with (NPEN) also easily satisfies **Why**, **Ontic and Factive**. It answers the question “*Why* is *p* necessary?”, by appealing to the *facts* about essence of things, not our concept of essence. Essentialists would certainly hold that if the essentialist explanation on necessity is valid, then it is **objectively** valid. Finally, **Structural** is not a negative constraint that one can break.

However, whether essentialism with (NPEN) is compatible with **Acyclicity** is perhaps not so obvious. Some might feel uncomfortable to the idea that the nature of to be necessary validates the explanation from essence to necessity. But if it is indeed circular, what exact part of the picture contains a circle?

Notice that I do not try to explain the necessity of *p* by (NPEN). The explanatory work is done by the fact that *p* is objectually essential. The urge to make (NP) explicit arises only when essentialists are asked how *all* necessities are explained by essence. Otherwise, the nature of to be necessary is enough to make the explanation valid.

Hence, there is at least no blatant circularity. Actually, it is even unclear why involving (NPEN) in particular explanations would be viciously circular. Suppose I am wrong in the previous paragraph, and every explanation on necessity must include (NPEN). p is necessary because p is essential to some objects *and every such truths are necessary in virtue of the nature of to be necessary*. What is wrong with this kind of explanation? My point can perhaps be made by a simple analogy. If someone asks me why 3 is a prime number, I can simply answer “because 3 is a natural number that is not a product of two natural numbers smaller than itself.” However, if someone thinks this is incomplete, then I can add: “... *and every such natural numbers are prime in virtue of the nature of to be prime*”. It would be obvious that neither explanation for the fact that 3 is prime is viciously circular.

More intriguing problems regarding circularity target specifically (NP): how is (NP) necessary? But it needs to be clarified whether essentialists are really committed to the necessity of (NP). I have formulated the account in a way that avoids such a commitment. Essentialists must hold that every fact essential to some *objects* is necessary. However, my proposal was that (NP) is the *predicational* essence of to be necessary, not an objectual essence of, for example the concept of necessity. Thus, (NPEN) does not force essentialists to admit that (NP) is necessarily true. Notice that by doing so one does *not* abandon the necessity of necessity: $\Box p \rightarrow \Box \Box p$. This is again because $E_F p \rightarrow E_F E_F p$ is a theorem where F is rigid.

However, there certainly are some circumstances where essentialists are automatically committed to the necessity of (NP). One obvious case is that if one insists the (NP) is true in virtue of the nature of the logical object of necessity. In this case (NP) itself commands (NP) to be necessary. Similar point can be made if one understands essence in terms of generalized identity, and prefers to take $\Box(p \equiv p)$ to be the partial objectual essence of necessity. This is the option considered in the end of the fourth section. If an essentialist prefers that option, by the necessity of generalized identity one would be committed to $\Box(\exists q(([\lambda G.Gx \equiv_x Nx]F) \equiv_F (\forall p \Box(p \equiv p) \wedge q)))$.

Or, some essentialists might wish to include (NPEN) in all explanation of necessity. Here the explanans of the necessity of p would be the conjunction of the essentiality of p and (NPEN). Since essentialists try to give a non-transmissive explanation, the conjunction should also be necessary, although the explanation does not appeal to the necessity. And since for a conjunction to be necessarily true each conjunct should be necessarily true, (NPEN) should be necessary. And it would be obvious that if (NPEN) is necessary, also is (NP).¹⁸

¹⁸ Actually, this is a disturbing consequence because it is *not* a theorem in the logic of essence

Now let us temporarily grant that (NP) is necessary, and therefore by essentialism (NP) is essential to some objects. Let such an object be n . Now, it does not pose a serious problem if (NP) is an essence of to be necessary independently of being the essence of n . But perhaps it is also plausible to suppose that (NP) being the essence of n grounds (NP) being the essence of to be necessary. Therefore, (NP) holds ultimately because of the nature of n , and my suggestion collapses to the thesis that the gap between essence and necessity is bridged by the nature of n .

The puzzle is then as follows. Essentialists will account for the source of $\Box(\text{NP})$ by $E_n(\text{NP})$. Now it may seem there is a gap between $E_n(\text{NP})$ and $\Box(\text{NP})$, which must be bridged by $E_n(\text{NP})$. That is: $E_n(\text{NP})$ grants $\Box(\text{NP})$ because $E_n(\text{NP})$. This is apparently circular, or amounts to say (NP) is *ab initio* necessary. And I admitted that objectual essence are not *ab initio* necessary.

One answer to the alleged puzzle is, of course, to deny (NP) is necessarily true. But even we grant the necessity of (NP), I think there is a possible reply. Recall the argument from $E_\alpha\Phi\alpha$ to $\Box\Phi\alpha$ in section 4. The essentiality of (NP) is indeed one crucial premises of the argument, but not the sole premise. The proof utilized T -axiom of the logic of essence and the inference rule modus ponens and universal instantiation. And it is quite customary to suppose the instances of logical axioms and inference rule is essential to the relevant logical notions (Fine 1994, 1995a). Therefore, one possible answer to the puzzle is to say, in the most rigorous sense, the answer to the problem of why $E_n(\text{NP})$ grants $\Box(\text{NP})$ should appeal to not only the essence of n but also the essentialist operator, conditional and universal quantifier.

One might feel uncomfortable to speak of the essence of essentialist operator E . For one thing, did not I reject to regard (NP) to be true in virtue of the essence of essence considering fundamentality? Yes, but no “modal axioms”, that is, axioms governing the essentialist operator involve notions on necessity or possibility. To say modal axioms are essential to essence does not result to the anti-essentialist consequence. It nevertheless does suggest that essence is ontologically dependent on logical connectives, set-theoretical notions, or some notions represented by the metalanguage of logic. This, however, is what essentialists *can* admit. Fine, for instance, clarifies himself as holding essence to be “methodologically primitive”, which is “perfectly compatible with its not actually being a primitive.” (Fine 2020, p.465)

that $\Box_F A \rightarrow \Box_F \Box_F A$, where F is a non-rigid predicate such as “to be necessary”. I think the suggestion that (NP) should be involved in source of all necessity goes better rather with the alternative that (NP) is true in virtue of some *objects* related to necessity. And that gives us a further reason to take (NP) to be necessary.

6 Conclusion

I argued that essentialists, facing the problem of gap, must respond by accepting (NPEN) - that (NP) is true in virtue of the nature of to be necessary. This follows from the more general character of non-transmissivism. Not only essentialist theories on the source of necessity, but all non-transmissivist theories face the problem of the gap. When one asserts that p is necessary because $C(p)$, one has to elucidate the notion of C without modal notions, and at the same time grant (GNP): $\forall p(C(p) \rightarrow \Box p)$. I argued in section 3 that all non-transmissivists should regard (GNP) to be a partial essence of to be necessary. And since all essentialists theories are non-transmissivist - this is argued in section 1 - essentialists are forced to accept (NPEN). I also tried to show that this account is not circular as it may first appear.

It is worth noting that (NPEN) seems to yield the conclusion that essentialists cannot hold the notion of to be necessary to be metaphysically fundamental. It should be ontologically dependent to the notion of to be essentially true of something. Therefore, if Hale meant metaphysical fundamentality by “fundamental and irreducible” character of necessity, his position seems no longer tenable. Either the essentialist theory on the source of necessity or the fundamentality of necessity must go.

What I did *not* argue in this paper is that essentialism is superior to its competitors. My aim in this paper has been to show that the particular challenge to essentialists can be met coherently without dismissing the challenge. Although what the challenge from gap points out is correct, it is not strong enough to defeat essentialism. It rather helps essentialists to clarify what comes together with essentialism. Therefore, one can read this conclusion as a conditional: *if* essentialism is correct, then so is (NPEN). Any argument in favor of essentialism is also an argument for (NPEN). But of course the contrapositive holds. Any criticism against (NPEN) would thereby be a criticism for essentialism. Dealing with those criticisms, however, would exceed the scope of this paper.

References

- Blackburn, S. (1993). Morals and modals. In S. Blackburn (Ed.), *Essays in quasi-realism* (pp. 52–73). Oxford University Press.
- Cameron, R. P. (2010). On the source of necessity. In B. Hale & A. Hoffmann (Eds.), *Modality: metaphysics, logic, and epistemology* (pp. 137–152). Oxford University Press.

- Correia, F. (2020). More on the reduction of necessity to essence. In M. Dumitru (Ed.), *Metaphysics, meaning, and modality: Themes from kit fine* (pp. 265–282). Oxford University Press.
- Correia, F., & Skiles, A. (2017). Grounding, essence, and identity. *Philosophy and Phenomenological Research*, 98(3), 642–670. doi: 10.1111/phpr.12468
- Fine, K. (1982). The problem of non-existents: I. internalism. *Topoi*, 1(1-2), 97–140. doi: 10.1007/bf00157548
- Fine, K. (1994). Essence and modality. *Philosophical Perspectives*, 8(Logic and Language), 1–16. doi: 10.2307/2214160
- Fine, K. (1995a). The logic of essence. *Journal of Philosophical Logic*, 24(3), 241–273. doi: 10.1007/bf01344203
- Fine, K. (1995b). Ontological dependence. *Proceedings of the Aristotelian Society*, 95(1), 269–290. doi: 10.1093/aristotelian/95.1.269
- Fine, K. (2012). Guide to ground. In F. Correia & B. Schnieder (Eds.), *Metaphysical grounding: understanding the structure of reality* (pp. 37–80). Cambridge University Press.
- Fine, K. (2015). Unified foundations for essence and ground. *Journal of the American Philosophical Association*, 1(2), 296–311. doi: 10.1017/apa.2014.26
- Fine, K. (2020). Comments on penelope mackie’s ”can metaphysical modality be based on essence?”. In M. Dumitru (Ed.), *Metaphysics, meaning, and modality: Themes from kit fine* (pp. 461–465). Oxford University Press.
- Glazier, M. (2017). Essentialist explanation. *Philosophical Studies*, 174(11), 2871–2889. doi: 10.1007/s11098-016-0815-z
- Hale, B. (2002). The source of necessity. *Noûs*, 36(s16), 299–319. doi: 10.1111/1468-0068.36.s16.11
- Hale, B. (2013). *Necessary beings: An essay on ontology, modality, and the relations between them*. Oxford, England: Oxford University Press.
- Leech, J. (2018). Essence and mere necessity. *Royal Institute of Philosophy Supplement*, 82, 309–332. doi: 10.1017/s1358246118000139
- Leech, J. (2021). From essence to necessity via identity. *Mind*, 130(519), 887–908. doi: 10.1093/mind/fzaa012
- Lowe, E. J. (2008). Two notions of being: Entity and essence. *Royal Institute of Philosophy Supplement*, 62, 23–48. doi: 10.1017/s1358246108000568
- Lowe, E. J. (2012). What is the source of our knowledge of modal truths? *Mind*, 121(484), 919–950. doi: 10.1093/mind/fzs122
- Mackie, P. (2020). Can metaphysical modality be based on essence? In M. Dumitru (Ed.), *Metaphysics, meaning, and modality: Themes from kit fine* (pp. 247–264). Oxford University Press.
- Priest, G. (2005). *Towards non-being: The logic and metaphysics of intention-*

- ality. New York: Oxford University Press.
- Rayo, A. (2013). *The construction of logical space*. Oxford, GB: Oxford University Press.
- Romero, C. (2019). Modality is not explainable by essence. *Philosophical Quarterly*, 69(274), 121–141. doi: 10.1093/pq/pqy043
- Romero, C. (2023). On explaining necessity by the essence of essence. *Inquiry: An Interdisciplinary Journal of Philosophy*, 1–24. doi: 10.1080/0020174x.2023.2186948
- Schaffer, J. (2009). On what grounds what. In D. Chalmers, D. Manley, & R. Wasserman (Eds.), *Metametaphysics: New essays on the foundations of ontology* (pp. 347–383). Oxford University Press.
- Wallner, M., & Vaidya, A. (2020). Essence, explanation, and modality. *Philosophy*, 95(4), 419–445. doi: 10.1017/s0031819120000200
- Wildman, N. (2018). Against the reduction of modality to essence. *Synthese*, 198(S6), 1455–1471. doi: 10.1007/s11229-017-1667-6
- Wilsch, T. (2017). Sophisticated modal primitivism. *Philosophical Issues*, 27(1), 428–448. doi: 10.1111/phs.12100
- Zalta, E. N. (1983). *Abstract objects: An introduction to axiomatic metaphysics*. Dordrecht, Netherland: D. Reidel.
- Zalta, E. N. (2006). Essence and modality. *Mind*, 115(459), 659–693. doi: 10.1093/mind/fzl659